Category
Equity Valuation: Concepts and Basic Tools

Related Questions:
Practice questions related to this topic

METHOD OF COMPARABLES AND VALUATION BASED ON PRICE MULTIPLES

Learning Outcomes

- explain the rationale for using price multiples to value equity, how the price to earnings multiple relates to fundamentals, and the use of multiples based on comparables
- calculate and interpret the following multiples: price to earnings, price to an estimate of operating cash flow, price to sales, and price to book value
- explain advantages and disadvantages of each category of valuation model

The method of comparables is the most widely used approach for analysts reporting valuation judgments on the basis of price multiples. This method essentially compares relative values estimated using multiples or the relative values of multiples. The economic rationale underlying the method of comparables is the **law of one price**: Identical assets should sell for the same price. The methodology involves using a price multiple to evaluate whether an asset is fairly valued, undervalued, or overvalued in relation to a benchmark value of the multiple. Choices for the benchmark multiple include the multiple of a closely matched individual stock or the average or median value of the multiple for the stock's industry. Some analysts perform trend or time-series analyses and use past or average values of a price multiple as a benchmark.

Identifying individual companies or even an industry as the "comparable" may present a challenge. Many large corporations operate in several lines of business, so the scale and scope of their operations can vary significantly. When identifying comparables (sometimes referred to as "comps"), the analyst should be careful to identify companies that are most similar according to a number of dimensions. These dimensions include (but are not limited to) overall size, product lines, and growth rate. The type of analysis shown in Section 5.1 relating multiples to fundamentals is a productive way to identify the fundamental variables that should be taken into account in identifying comparables.

EXAMPLE 13

Method of Comparables (1)

1. As noted previously, P/E is a price multiple frequently used by analysts. Using P/E in the method of comparables can be problematic, however, as a result of business cycle effects on EPS. An alternative valuation tool that is useful during periods of economic slowdown or extraordinary growth is the P/S multiple. Although sales will decline during a recession and increase during a period of economic growth, the change in sales will be less than the change in earnings in percentage terms because earnings are heavily influenced by fixed operating and financing costs (operating and financial leverage).

   The following data provide the P/S for most of the major automobile manufacturers as at December 2017:

Company	P/S
Peugeot	0.28
Ford Motor	0.33
General Motors	0.36
Nissan Motor	0.38
Honda Motor	0.46

BMW	0.57
Toyota Motor	0.80

Sources: Morningstar and company websites.

Based on the data presented, which stock appears to be undervalued when compared with the others?

Hide Solution

Solution:

The P/S analysis suggests that Peugeot shares offer the best value. An analyst must be alert for a range of potential explanations of apparently low or high multiples when performing comparables analysis, rather than just assuming a relative mispricing.

EXAMPLE 14

Method of Comparables (2)

1. Incorporated in the Netherlands, Airbus is active in the aerospace and defense industry. It is a dominant aerospace company in Europe. Its largest business, Airbus Commercial Aircraft, is a manufacturing company with bases in several European countries and accounts for the majority of Airbus SE profits. Airbus and its primary competitor, Boeing, control most of the global commercial airplane industry.

   Comparisons are frequently made between Airbus and Boeing. As noted in Exhibit 7, the companies are broadly similar in size as measured by total revenues. Converting total forecast revenues from euros to US dollars using the average exchange rate for 2017 of US$1.13/€ results in a value of $75.5 billion for Airbus's total revenues. Thus, total revenues for Boeing are expected to be about a fifth higher than those for Airbus.

   The companies do differ, however, in several important areas. Airbus derives a greater share of its revenue from commercial aircraft production than does Boeing, and the order backlog for Airbus is much higher than that for Boeing. Converting the Airbus order backlog from euros to US dollars using the quarter-end rate for September 2017 of $1.1813/€ results in a value of $1.12 billion for Airbus's order backlog. Thus, the order backlog for Airbus is more than twice as high as the backlog for Boeing.[5]

Exhibit 7:

Data for EADS and Boeing

	Airbus	Boeing
Total revenues (billions, 2017)	€66.8	$92.2
Annual revenue growth (2015–2017 average)	1.8%	–2.1%
Percent of revenues from commercial aircraft	75%	69%
Order backlog (billions)	€945	$474
Share price, 12/Dec/17	€86.96	$283.73
EPS (basic)	€3.33	$10.18
DPS	€1.48	$5.7
Dividend payout ratio	44%	56%
P/E ratio	26.1	27.9

What data shown in Exhibit 7 support a higher P/E for Boeing than for Airbus?

Hide Solution

Solution:

Recall from Equation 14 and the discussion that followed it that P/E is directly related to the payout ratio and the dividend growth rate. The P/E is inversely related to the required rate of return. The only data presented in Exhibit 7 that support a higher P/E for Boeing is the company's higher dividend payout ratio (expected at 56 percent versus 44 percent for Airbus).

The following implicitly supports a higher P/E for Airbus: Airbus has higher revenue growth (as reported for 2016 and expected for 2017) and a higher backlog of orders, suggesting that it may have a higher future growth rate.

EXAMPLE 15

Method of Comparables (3)

1. Canon Inc. is a leading worldwide manufacturer of business machines, cameras, and optical products. Canon was founded in 1937 as a camera manufacturer and is incorporated in Tokyo. The corporate philosophy of Canon is *kyosei* or "living and working together for the common good." The following data can be used to determine a P/E for Canon over the time period 2013–2017. Analyze the P/E of Canon over time and discuss the valuation of Canon.

Year	Price (a)	EPS (b)	P/E (a) ÷ (b)
2013	¥3,330	¥200.8	16.6
2014	¥3,840.5	¥229.0	16.8
2015	¥3,675	¥201.7	18.2
2016	¥3,295	¥138	23.9
2017	¥4,200	¥222.88	18.8

Sources: EPS, year-end prices, and P/E data are from Capital IQ and the *Financial Times*.

Hide Solution

Solution:

Trend analysis of Canon's P/E reveals a peak of 23.9 at the end of 2016. The 2013 P/E of 16.6 is the lowest of the five years reported. This finding suggests that Canon's share price may be fairly price as of year-end 2017. A bearish case for Canon's stock can be made if an analyst believes that P/E will return to its historical low (16.6 over this five-year period) or be lower. Such a bearish prediction requires that a decrease in P/E not be offset by an increase in EPS. A bullish case can be made if the analyst believes the stock deserves re-rating and an even higher than trend P/E.

Illustration of a Valuation Based on Price Multiples

Telefónica S.A., a world leader in the telecommunication sector, provides communication, information, and entertainment products and services in Europe, Africa, and Latin America. It has operated in its home country of Spain since 1924, but as of 2017, more than 75 percent of its business was outside its home market.

Deutsche Telekom AG provides network access, communication services, and value-added services via fixed and mobile networks. It generates more than half of its revenues

Rate Your Confidence

Category
Equity Valuation: Concepts and Basic Tools

Related Questions:
Practice questions related to this topic

Exhibit 8:

Data for Telefónica and Deutsche Telekom

	Telefónica			Deutsche Telekom		
	2017	2016	2015	2017	2016	2015
(1) Total assets (€ billions)	115.0	123.6	120.3	141.3	148.5	143.9
Asset growth	−6.9%	2.7%	—	−4.9%	3.2%	—
(2) Net revenues (€ billions)	52.0	52.0	54.9	77.3	75.2	71.3
Revenue growth	0%	−5.2%	—	2.8%	5.5%	—
(3) Net cash flow from operating activities (€ billions)	13.8	13.3	13.6	17.2	15.5	15.0
Cash flow growth	3.4%	−2.0%	—	11.0%	3.3%	—
(4) Book value of common shareholders' equity (€ billions)	16.9	18.2	15.8	42.5	38.8	38.2
Debt ratio: 1 – [(4) ÷ (1)]	85.3%	85.3%	86.9%	70.0%	73.9%	73.5%
(5) Net profit (€ billions)	2.9	2.1	0.4	3.5	2.7	3.3
Earnings growth	38.1%	425.0%	—	29.6%	−18.2%	—
(6) Weighted average number of shares outstanding (millions)	5,122.9	4,896.6	4,833.6	4,740.2	4,654.9	4,584.8
(7) Price per share (€)	7.93	9.52	9.13	13.42	16.21	15.44
Price-to-revenue ratio (P/R): (7) ÷ [(2) ÷ (6)]	0.8	0.9	0.8	0.8	1.0	1.0
P/CF: (7) ÷ [(3) ÷ (6)]	2.9	3.5	3.2	3.7	4.9	4.7
P/B: (7) ÷ [(4) ÷ (6)]	2.4	2.6	2.8	1.5	1.9	1.9
P/E: (7) ÷ [(5) ÷ (6)]	14.0	22.2	110.3	18.2	27.9	21.5

Sources: Company websites: www.telefonica.es and www.deutschetelekom.com.

Time-series analysis of all price multiples in Exhibit 8 suggests that both companies are currently attractively valued. For example, the 2017 price-to-revenue ratio (P/R) of 0.78 for Telefónica is below the 2015–2017 average for this ratio of approximately 0.83. The 2017 P/CF of 3.7 for Deutsche Telekom is below the 2015–2017 average for this ratio of approximately 4.4.

A comparative analysis produces somewhat mixed results. The 2017 values for Deutsche Telekom for the P/R, P/CF, P/E multiples are higher than those for Telefónica. This result suggests that Telefónica is attractively valued when compared with Deutsche Telekom. The 2017 P/B for Telefónica, however, is higher than for Deutsche Telekom.

An analyst investigating these contradictory results would look for information not reported in Exhibit 8. For example, the earnings before interest, taxes, depreciation, and amortization (EBITDA) for Telefónica was €16.4 billion in 2017. The EBITDA value for Deutsche Telekom was €20.7 billion in 2017. The 2017 price-to-EBITDA ratio for Telefónica is [(7.93 × 5,123)/16,400] or [7.92/(16,400/5,123] = 2.5, whereas the 2017 price-to-EBITDA ratio for Deutsche Telekom is 3.1. Thus, the higher P/E for Deutsche Telekom can-not be explained by higher depreciation charges, higher interest costs, and/or a greater tax burden, but appears to be due to a better quality of earnings.

In summary, the major advantage of using price multiples is that they allow for relative comparisons, both cross-sectional (versus the market or another comparable) and in time series. The approach can be especially beneficial for analysts who are assigned to a particular industry or sector and need to identify the expected best performing stocks within that sector. Price multiples are popular with investors because the multiples can be

benchmarks but overvalued when compared with an estimate of intrinsic value as determined by one of the discounted cash flow methodologies. Furthermore, differences in reporting rules among different markets and in chosen accounting methods can result in revenues, earnings, book values, and cash flows that are not easily comparable. These differences can, in turn, result in multiples that are not easily comparable. Finally, the multiples for cyclical companies may be highly influenced by current economic conditions.

Rate Your Confidence

Discuss

Filter

Discussion

What do you want to discuss... (required)

No messages to display

© 2025 CFA Institute. All Rights Reserved.

Rate Your Confidence

Category

Equity Valuation: Concepts and Basic Tools

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Equity Valuation: Concepts and Basic Tools

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Equity Valuation: Concepts and Basic Tools

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Equity Valuation: Concepts and Basic Tools

Related Questions:

Practice questions related to this topic